



LKMamba-CD: Large Kernel State Space Model for Remote Sensing Change Detection

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Abstract

Remote sensing change detection (RSCD) requires effective modeling of long-range spatial dependencies and large receptive fields to accurately capture subtle land-cover variations between bi-temporal remote sensing images. However, existing RSCD methods often suffer from limited receptive fields or high computational complexity, constraining their ability to capture large-scale contextual dependencies. To address these challenges, this paper proposes a Large Kernel State Space Model for Change Detection (LKMamba-CD). Within LKMamba-CD, the Large Kernel SS2D Module (LKS) is designed to extract large-scale and long-range feature correlations. The Bi-temporal Change Information Fusion Module (BCIF) is utilized to fuse bi-temporal change information. Meanwhile, the Large Receptive Field Feature Aggregation Module (LRFA) is utilized to extract large-scale land cover feature correlation. Within BCIF and LRFA, LKS serves as the core component, leveraging re-parameterized large kernel convolution and the Mamba architecture to capture long-range dependencies with expansive receptive fields efficiently. Experiments on S2Looking, WHU-CD, and GVLN-CD show that LKMamba achieves the highest F1 of 62.02%, 92.52%, and 87.58%, respectively. The code of LKMamba-CD at <https://github.com/kion-86/lkmamba>.

Keywords Remote sensing change detection · Re-parameterized large kernel convolution · State Space Model · Large receptive-fields

1 Introduction

Remote sensing change detection (RSCD) encompasses a broad spectrum of techniques aimed at identifying Earth's surface dynamics by analyzing remote sensing images (RSIs) acquired over multiple time points Wang et al. (2023). Beyond the foundational paradigm of optical bi-

temporal image comparison, RSCD has evolved into highly sophisticated multi-temporal frameworks. Particularly in the domain of Synthetic Aperture Radar (SAR), Coherent Change Detection methods leverage phase information and temporal decorrelation models to detect subtle environmental variations Jung et al. (2016). These techniques utilize physical coherence matrices across SAR acquisitions rather than being limited to bi-temporal pairs Monti-Guarnieri et al. (2018); Costa et al. (2024). Furthermore, contemporary deep learning have been uniquely tailored to jointly process these multi-temporal stacks, resulting in physics-aware architectures that significantly enhance monitoring capabilities Costa et al. (2025a, b). While advanced multi-pass methodologies drive the frontiers of SAR monitoring, optical bi-temporal RSCD remains a critical and highly active sub-field, playing a crucial role in monitoring dynamic environmental evolution of the Earth's surface. It has been widely applied in diverse domains such as urban planning, land-use monitoring, disaster assessment, and ecological management Dai et al. (2023).

Traditional RSCD methods, such as Principal Component Analysis and Multivariate Alteration Detection

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Nielsen et al. (1998), mainly rely on spectral information and statistical transformations to extract change features. However, these methods often require manual threshold selection during the decision-making stage, leading to limited generalization ability, high dependence on specific scenarios, and increased computational costs Zhu et al. (2025). To overcome these limitations, machine learning-based approaches, including Support Vector Machines Negri et al. (2020) and Random Forests Feng et al. (2018), have been introduced to automatically learn discriminative change representations from limited labeled samples. Although these approaches enhance automation and reduce manual intervention, their performance heavily depends on handcrafted features Guo et al. (2022). Consequently, they struggle to capture the complex, high-level semantic relationships inherent in RSIs, which restricts their detection accuracy and robustness Ren et al. (2024a). With the development of deep learning (DL), Convolutional Neural Networks (CNNs) show significant advances in RSCD tasks Caye Daudt et al. (2018) due to their strong feature representation capability. By stacking convolutional layers, CNNs can effectively extract hierarchical pixel-level features. Nevertheless, the fixed and limited receptive field of CNNs constrains their ability to model long-range spatial dependencies Chen et al. (2025). In RSIs, land-cover structures often span large spatial extents, making CNNs less effective in capturing global contextual information Ren et al. (2024a). To address this limitation, RepLKNet Ding et al. (2022) employs re-parameterized large-kernel convolutions to expand the receptive field, thereby improving the modeling of large-scale spatial patterns. Building upon this, CD-RLKNet Huang et al. (2024) has demonstrated the effectiveness of large-kernel convolutions for change detection. However, these methods remain inherently limited in capturing global dependencies among remote sensing features, as their representational power is still confined to local spatial regions.

To capture true global dependencies, Transformers leveraging self-attention mechanism, have achieved powerful long-range modeling capabilities, enabling them to capture global dependencies in RSIs Vaswani et al. (2017). However, the computational complexity of the self-attention mechanism within Transformer exhibits a quadratic relationship. As image sizes grow, computational costs increase rapidly, constraining their applicability in RSCD tasks Fang et al. (2022); Huang et al. (2026). This computational bottleneck motivates the exploration of more efficient architectures to balance global context modeling with scalability. State Space Models (SSM) Gu et al. (2021), as a new paradigm for long-sequence modeling, offer linear computational complexity while maintaining strong global modeling capacity. Mamba, a recent variant of SSM, has achieved outstanding performance across computer visual

tasks Liu et al. (2024). RS-Mamba Zhao et al. (2024a) extends Mamba to RSCD by capturing long-range dependencies from multiple directions. However, Mamba's reliance on global aggregation tends to weaken pixel-level spatial precision, as it sacrifices local spatial details for global modeling efficiency. In pixel-wise dense prediction tasks like RSCD, this trade-off significantly constrains performance Huang et al. (2025).

To address these challenges, the Large Kernel State Space Model for Change Detection (LKMamba-CD) is proposed. LKMamba-CD combines re-parameterized large kernel convolution with the Mamba to aggregate the correlation of long-range remote sensing features within a large receptive field. LKMamba-CD utilizes UniRepLKNet as the backbone to extract land cover features from RSIs. Meanwhile, the Large Kernel SS2D Module (LKS) is designed as a core component, which combine re-parameterized large kernel convolution with SS2D, to achieve large-scale, long-range pixel-level feature correlation modeling while maintaining linear complexity. To integrate bi-temporal land cover features, Bi-temporal Change Information Fusion Module (BCIF) is designed based on LKS to extract the difference information of land cover features at different temporal, thereby integrating the change information of multi-dimensional spatio-temporal features. The Large Receptive-field Feature Aggregation Module (LRFA) uses efficient attention and LKS to strengthen the correlation of change features in different dimensions through large-scale feature extraction, allowing for the preservation of high-dimensional features without losing large-scale change information across different dimensions. The proposed LKMamba-CD effectively aggregates long-range spatial dependencies within a large receptive field while maintaining linear computational complexity, achieving an optimal balance between local feature preservation and global dependency modeling. The contributions are as follows.

- We propose a LKMamba-CD that seamlessly integrates large-kernel convolutions and the Mamba, combining the local spatial representation strength of CNNs with the efficient long-range modeling capability of Mamba.
- LKS is designed by coupling re-parameterized large-kernel convolutions with two-dimensional state-space modeling (SS2D), enabling large-scale, pixel-level dependency aggregation with linear complexity.
- BCIF is introduced in RSCD to extract and fuse multi-dimensional difference information. This operation enhances cross-temporal semantic alignment, thereby effectively integrating bi-temporal land cover features for accurate change detection.
- LRFA is designed to leverage efficient attention mechanisms and LKS to strengthen multi—dimensional change

feature correlations, preserving high-level semantics while enhancing large-scale variation perception.

2 Related Work

Deep learning-based RSCD approaches primarily employ CNNs to automatically learn discriminative feature representations from RSIs, facilitating the extraction of complex spatial patterns and contextual dependencies among pixels. With the growing success of CNNs in computer vision, researchers have increasingly adopted these architectures for RSCD tasks. Daudt et al. (2018) were the first to introduce FCN Long et al. (2015) into RSCD by proposing FC-Siam-diff, a siamese CNNs model that extracts spatial features from bi-temporal RSIs to detect surface changes. Subsequently, Fang et al. (2022) presented SNUNet, which enhances deep spatio-temporal feature fusion through dense connections, improving the precision of change detection. Huang et al. (2024b) proposed F3Net, which integrates multi-scale feature fusion mechanisms within CNNs to filter out noise and strengthen change representations across different resolutions. Despite their effectiveness, CNNs-based RSCD methods suffer from the intrinsic limitation of local receptive fields, which hinders their ability to model long-range spatial dependencies Zhao et al. (2024b). As a result, CNNs may fail to capture global contextual cues in RSIs, particularly in large-scale or complex geographical regions.

To overcome the intrinsic limitations of local receptive fields in CNNs, researchers have recently extended Transformers Vaswani et al. (2017), originally successful in natural language processing, to computer vision tasks Liu et al. (2021). Transformers utilize self-attention mechanisms to model long-range dependencies, allowing the network to focus adaptively on informative spatial regions within RSIs Zhang et al. (2022).

Several Transformer-based RSCD models have demonstrated notable progress. DSAMNet Shi et al. (2021) integrates spatial and channel attention mechanisms to improve feature discrimination and change localization. ChangeFormer Bandara and Patel (2022) employs a pure Transformer siamese architecture that dispenses with CNNs backbones entirely, achieving superior accuracy through hierarchical attention-based feature extraction. BIT Hao Chen and Shi (2021) reformulates bi-temporal RSIs into semantic tokens and models spatio-temporal relationships through a token-based representation, thereby improving the semantic consistency of detected changes. Moreover, ICIFNet Feng et al. (2022) employs hierarchical Transformers to capture intra-scale and inter-scale dependencies among bi-temporal features, further enhancing cross-temporal modeling capabilities. Although Transformer-based

models excel in capturing global dependencies, their self-attention mechanism incurs quadratic computational complexity with respect to image size Dao and Gu (2024). This restricts their scalability and efficiency in processing high-resolution RSIs, motivating research into more computationally efficient long-range modeling paradigms.

Parallel to the development of Transformers, exploring universal feature representations with expansive receptive fields has led to the resurgence of Large Kernel Convolutions (LKC). This necessity requires algorithms capable of capturing features from large-scale land cover, which in turn requires a model with an expansive receptive field Ren et al. (2024b). Today, researchers have explored the use of LKC to equip models with this capability. LKC expands on traditional CNNs kernels by increasing their size, thereby enhancing the ability to capture larger receptive fields and more effectively extract image features Lau et al. (2024).

To improve the performance of LKC, Liu et al. (2022) introduced ConvNeXt, a model that relies solely on standard CNN architectures using 7×7 convolutions, eschewing transformer structures while still achieving competitive results in terms of accuracy, scalability, and robustness. Their experiments demonstrated that modifying the size of the convolution kernel can significantly boost the performance of the model. Further advancements in this area include RepLKNet Ding et al. (2022), which adopts extremely large kernels (31×31) and leverages re-parameterizable large depth-wise separable convolutions to design high-performance CNNs. The findings indicate that, compared to smaller kernels, LKC not only provides a larger effective receptive field, but also exhibits higher shape bias, highlighting its advantages for downstream computer vision tasks. Moreover, Ding et al. (2023) proposed UniRepLKNet, a versatile perception model demonstrating exceptional performance in various domains such as audio, video, point cloud, time series, and image recognition. Utilizing SE Blocks to enhance depth and Dilated Reparam Blocks for convolutional layer reparameterization, among other techniques, UniRepLKNet showcases the robust capabilities of large kernel convolutions. This work strongly supports the application of LKC in remote sensing image processing, indicating significant potential for future developments in this field.

While large kernels expand the receptive field, SSM Gu et al. (2021) have emerged as a novel linear time-invariant system that fundamentally addresses the computational complexity of long-range sequence modeling.

SSM is introduced to computer vision and aims to address the computational complexity issues faced by traditional models when processing long-range sequence data. Mamba, Gu and Dao (2023); Dao and Gu (2024) as a sequence modeling method based on SSM with linear time complexity, effectively reduces computational complexity

through a selective scanning mechanism while maintaining efficient context modeling capabilities. VMamba successfully applies SSM to visual tasks by introducing two-dimensional state-space model (SS2D) Liu et al. (2024), achieving efficient image processing. VMamba inherits Mamba's global modeling capabilities and linear time complexity, while addressing the two-dimensional non-sequential nature of image data through the SS2D.

SS2D can effectively traverse spatial features using a four-directional scanning mechanism. Compared with self-attention, SS2D ensures that each image patch only obtains contextual information through the compressed hidden states calculated along the corresponding scan path. This enables VMamba to effectively capture global context information in images while maintaining efficient computation. VMamba is used in classification, object detection, and semantic segmentation, and has made significant progress in the remote sensing image process. Many scholars have also applied Mamba to RSCD. RS-Mamba Zhao et al. (2024a) utilizes OSSM to scan features from multiple directions, capturing large spatial features in remote sensing images. In contrast to RS-Mamba, LCCDMamba Huang et al. (2025) designs DTMS, which enhances feature correlations across different dimensions through two distinct modeling approaches. DTMS fully aggregates spatio-temporal features during the feature restoration process, preserves high-dimensional semantic features, and ensures that multi-scale change information is not lost throughout this process.

3 Methodology

3.1 Preliminaries

SSM is a linear time-invariant system that transforms a 1-dimensional function or sequence $x(t) \in \mathbb{R}$ into $y(t) \in \mathbb{R}$ via a hidden state $h(t) \in \mathbb{R}^N$. The system dynamics is captured by a linear ordinary differential equation (ODE) Gu et al. (2021):

$$h'(t) = Ah(t) + Bx(t) \quad (1)$$

$$y(t) = Ch(t) \quad (2)$$

where, N denotes the state size, with $A \in \mathbb{R}^{N \times N}$, $B \in \mathbb{R}^{N \times 1}$, and $C \in \mathbb{R}^{1 \times N}$.

To integrate continuous systems into DL algorithms, the S4 model introduces a time scale parameter Δ . This facilitates the discretization of the continuous parameters A and B using the zero-order hold (ZOH) method Gu and Dao (2023), producing discrete parameters $\bar{A}$ and $\bar{B}$:

$$\bar{A} = e^{\Delta A} \quad (3)$$

$$\bar{B} = (\Delta A)^{-1}(e^{\Delta A} - I)\Delta B \quad (4)$$

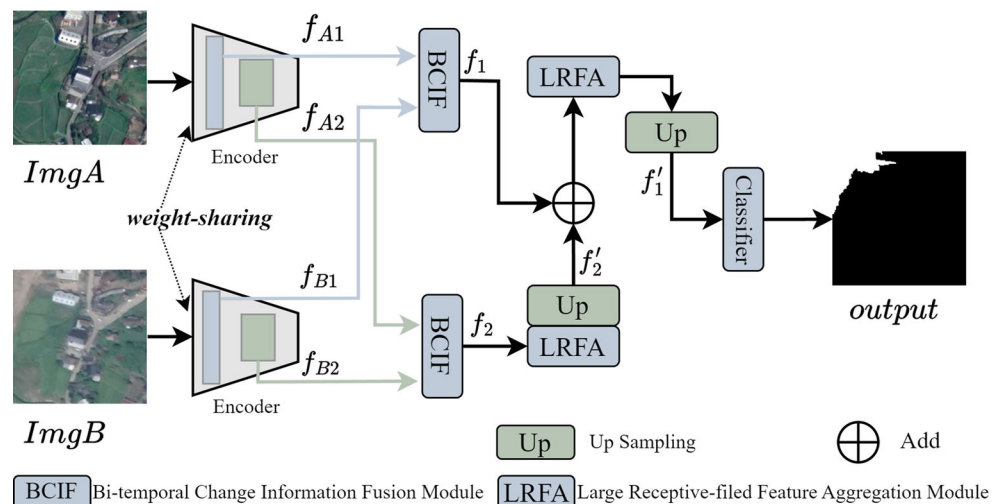
Post-discretization, the ODEs transform into:

$$h_t = \bar{A}h_{t-1} + \bar{B}x_t \quad (5)$$

$$y_t = Ch_t \quad (6)$$

On this basis, Mamba designed a selection mechanism to sequentially filter out irrelevant information from the model. Moreover, Mamba has also developed a known hardware-aware algorithm to accelerate the computation speed of the model on GPU, ensuring that Mamba can maintain linear increases in memory and computational costs.

Fig. 1 The structure of LK-Mamba-CD. LKMamba-CD using Siam-UniRepLKNet as Encoder to extract the difference information of land cover features at different temporal, thereby integrating the change information of multi-dimensional spatio-temporal features. LRFA is utilized to strengthen the correlation of change features in different dimensions through large-scale feature extraction



3.2 Overview

The structure of LKMamba-CD is illustrated in Fig. 1. To effectively capture multi-scale land cover variations in RSIs, which often exhibit complex spatial hierarchies and discrepancies, LKMamba-CD employs the dual branch Unireplknet Ding et al. (2023) with sharing weights as the Encoder. This Siamese architecture is particularly advantageous for change detection tasks, as it enables shared weights between branches to extract consistent and discriminative features from bi-temporal RSIs $ImgA$ and $ImgB$, yielding f_{A1} , f_{A2} , f_{B1} , and f_{B2} . Specifically, f_{A1} and f_{B1} are downsampled to 1/8 of the input image size, preserving mid-level details such as edges and textures, while f_{A2} and f_{B2} are further reduced to 1/32, facilitating the capture of high-level semantic information essential for distinguishing subtle land cover changes.

During the bi-temporal feature fusion stage and the feature recovery stage, to obtain a larger receptive field and long-range feature information, Large Kernel SS2D (LKS) is designed by combining re-parameterized large kernel convolution and Mamba. The Bi-temporal Change Information Fusion Module (BCIF) used LKS to integrate the cross-temporal differences between these features, producing fused representations f_1 and f_2 . BCIF enhances the

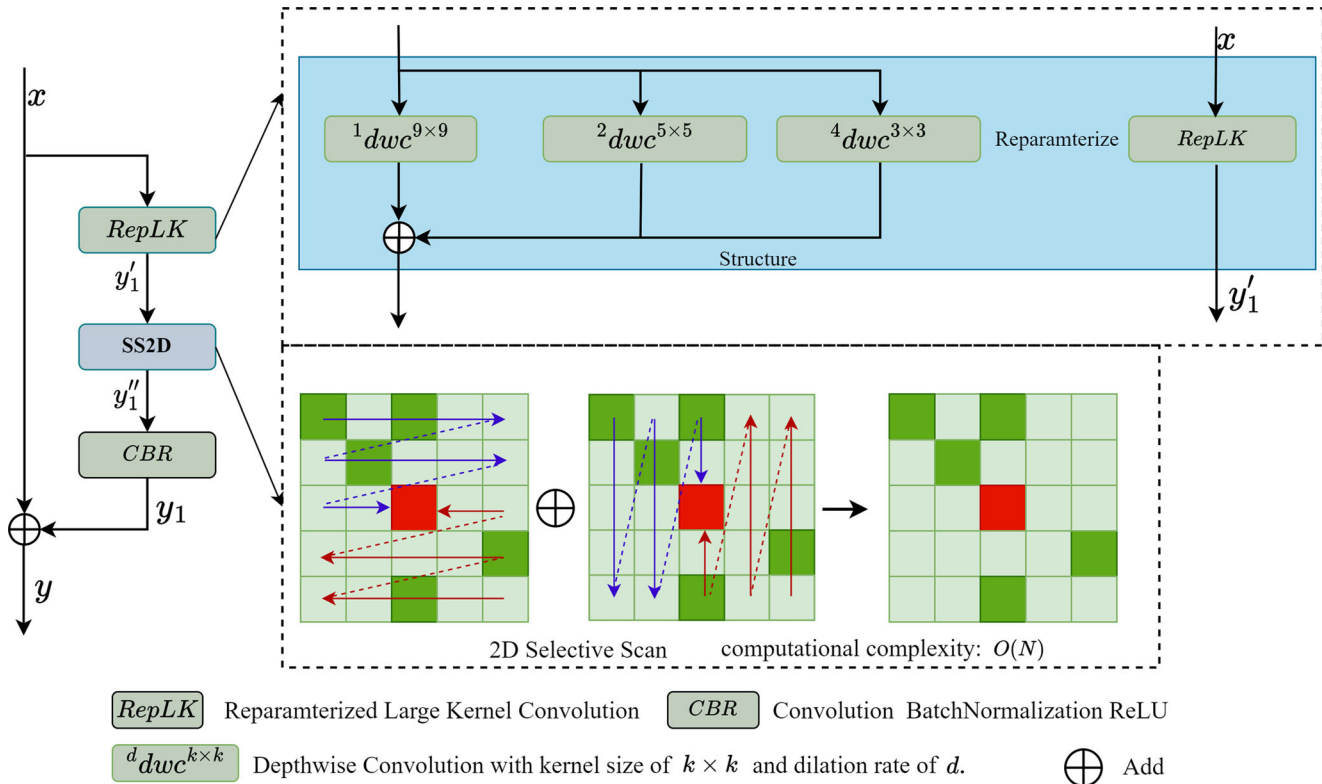
model's robustness by explicitly modeling inter-temporal interactions, mitigating issues like pseudo-changes, thereby improving the precision of change localization. Next, the Large Receptive-field Feature Aggregation Module (LRFA) performs feature recovery, leveraging its expansive receptive fields to aggregate contextual information across scales. This design is critical for RSCD applications, where large-scale spatial dependencies must be holistically considered to avoid fragmented detections. During the bi-temporal feature fusion stage and the feature recovery stage, to obtain a larger receptive field and long-range feature information, Large Kernel SS2D (LKS) is designed by combining re-parameterized large kernel convolution and Mamba. The feature recovery process is formalized as follow:

$$f'_2 = up(LRFA(f_2)) \quad (7)$$

$$f'_1 = up(LRFA(f_1 + f'_2)) \quad (8)$$

$$output = Conv^{7 \times 7}(f'_1) \quad (9)$$

where up denotes bilinear upsampling, and $Conv^{7 \times 7}$ indicates convolution with kernels of 7×7 . This large-kernel convolution serves as the classifier to refine boundary details and suppress noise in the output prediction, ensuring high-fidelity change maps.



3.3 Large Kernel SS2D Module

LKS is the key component of BCIF and LRFA, responsible for further spatial modeling of feature maps across different temporal and dimension. Thanks to the combination of re-parameterized large kernel convolution and SS2D, LKS can simultaneously achieve multi-scale, long-range pixel-level feature correlation modeling while maintaining linear complexity. This ensures that LKS outperforms previous methods that use CNNs for fusing local pixel-level information and Transformer for fusing long-range patch-level relationships. More importantly, LKS's low complexity and larger receptive field enhance the efficiency of local modeling. The structure of LKS is illustrated in Fig. 2.

For the input x of LKS, it first goes through re-parameterized large kernel convolution to obtain y'_1 .

$$y'_1 = \text{RepLK}(x) \quad (10)$$

where RepLK indicates re-parameterized large kernel convolution, RepLK applies depthwise convolution (DWC) with different dilation rates to capture land cover information at various scales. Through reparameterization, multiple DWC are integrated into a RepLK . Following the design principles of RepLKNet Ding et al. (2022), the receptive field of DWC is determined by convolution kernel size and the dilation rate, as described below.

$$k_{rf} = (k - 1) \times r + 1 \quad (11)$$

where k_{rf} indicates receptive field of DWC , k represents kernel size, r represents dilation rate.

Taking a 9×9 kernel as an example, the parameters k and r for each DWC are set to (9,1), (5,2), and (3,4), respectively. Different values of r will be adjusted for fusion. Finally, RepLK is obtained by re-parameterizing these three DWC s.

Re-parameterized convolution ensure that the receptive field of LKS can effectively capture the inductive biases inherent in remote sensing information, while avoiding the wastage of computational resources and the introduction of noisy features that may arise from large kernel sizes.

Next, SS2D is used to aggregate long-range feature correlations in y'_1 to obtain y'_1 . SS2D obtains spatio-temporal contextual information, effectively establishing a global receptive field and capturing rich contextual details. This is crucial for detecting subtle changes in complex scenes during RSCD tasks. The output y is calculated as follows.

$$y_1 = \rho(\text{bn}(\text{Conv}^{3 \times 3}(y''_1))) \quad (12)$$

$$y = y_1 + x \quad (13)$$

where $\text{Conv}^{3 \times 3}$ indicates convolution with kernel sizes of 3×3 . bn indicates BatchNormalization, ρ indicates ReLU.

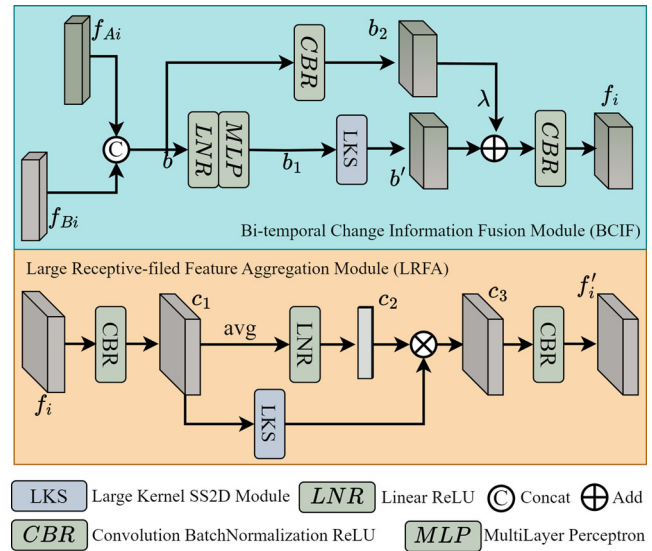


Fig. 3 The structure of BCIF and LRFA. BCIF fuse the difference information at same dimensions. LRFA integrates large-scale features correlation of change features across different dimensions

By combining RepLK and SS2D , the approach can effectively extract fine-grained feature correlations in RSIs. This integration maintains model complexity and generalization ability, thus enhancing the accuracy of RSCD tasks.

3.4 Bi-temporal Change Information Fusion Module

In LKMamba-CD, BCIF is used to aggregate the difference information of bi-temporal remote sensing features extracted by the Unireplknet. The structure of BCIF is illustrated in Fig. 3. In the BCIF process, the inputs f_{Ai} and f_{Bi} are initially concatenated to obtain b .

$$b = \text{concat}(f_{Ai}, f_{Bi}) \quad (14)$$

where concat indicates concatenate.

Next, b is processed through two separate branches. One branch undergoes linear mapping, followed by processing through LKS to obtain b' . The calculation process is as follows.

$$b_1 = \text{MLP}(\rho(\text{LN}(b))) \quad (15)$$

$$b' = \text{LKS}(b_1) \quad (16)$$

where LN indicates Linear, MLP indicates multilayer perceptron. In this process, the combination of LN and MLP is used to compress the channels of b , LN have a small computational requirement, while MLP can handle various types of information, allowing the transmission of features to LKS with minimal computation and higher efficiency.

In another branch, CNNs are utilized to compress the channels of b to obtain b_2 . After weighting and adding b'

and b_2 , the result of the BCIF output f_i is obtained by CNNs. The calculation process is as follows.

$$b'' = \lambda \times b_2 + b' \quad (17)$$

$$f_i = \rho(bn(Conv^{1 \times 1}(b''))) \quad (18)$$

Here, a learnable parameter λ is utilized as the coefficient for b_2 , which serves to balance the mutual influence between the deep features aggregated by LKS and the basic features.

3.5 Large Receptive-filed Feature Aggregation Module

After the encoder extracts land cover features, BCIF is utilized to fuse land cover information from different temporal and aggregate change information. For restoring fine-grained change features from the multi-dimensional change information, ultimately yielding highly precise detection results. LRFA is designed to recover change features, which utilizes efficient attention and LKS to fuse change features f_1 and f_2 . LRFA is illustrated in Fig. 3.

LRFA takes high-dimensional features as input. Initially, CNNs is applied to compress channels of feature map. This is followed by average pooling and a linear transformation to obtain c_2 . The calculation process is as follows.

$$c_1 = \rho(bn(Conv^{1 \times 1}(f_i))) \quad (19)$$

$$c_2 = \rho(LN(avg(c_1))) \quad (20)$$

where avg indicates average pooling.

Next, LKS is used to further capture the multi-scale feature correlations of c_1 , and multiplied by c_2 to obtain c_3 .

$$c_3 = c_2 \times LKS(c_1) \quad (21)$$

Finally, the output of LRFA is obtained by a convolution calculation.

$$f'_i = \rho(bn(Conv^{3 \times 3}(c_3))) \quad (22)$$

In LRFA, LKS learns the spatio-temporal relation within different aspects of changing features from a larger receptive field, capturing the intrinsic connections of change information derived from spatio-temporal long-range. Simultaneously, an efficient attention mechanism is utilized to effectively aggregate the changing features of the land cover.

4 Experiments

4.1 Dataset Description

S2Looking Shen et al. (2021) is a dataset dedicated to detecting changes that covers 5000 pairs of RSIs from rural areas around the world. The dataset includes over 65,920 annotated change instances. To ensure complete independence in the evaluation and avoid bias introduced by spatial correlation, the *S2Looking* dataset was partitioned into training, validation, and test sets at a ratio of 7:1:2 by the original authors. There is no overlap among the images in the three sets.

WHU-CD Ji et al. (2018) contains RSIs taken in New Zealand in April 2012 and April 2016, with image dimensions of 32507×15354 pixels. The spatial resolution achieves a fine detail of 0.2m per pixel, covering a geographical area of approximately 20.5 square kilometers. To prevent data leakage and spatial autocorrelation, we partitioned the image into non-overlapping spatial regions, yielding 7400 pairs of 256×256 image pairs. These are split into training, validation, and test sets at a ratio of 7:2:1. There is no overlap among the images in the three sets.

GVLM-CD Zhang et al. (2023b) contains 17 pairs of bi-temporal, imagery with a spatial resolution of 0.59m, acquired via Google Earth. The total coverage of the dataset is 163.77 km^2 . To ensure complete data independence, the dataset is partitioned at the scene level rather than the patch level. The entire dataset yields 7600 pairs of 256×256 image pairs cropped by scene. These are split into training, validation, and test sets at a ratio of 7:2:1. There is no overlap among the images in the three sets.

4.2 Implementation Details

LKMamba-CD is implemented using PyTorch and trained on RTX4090D. The AdamW optimizer is used for training, with an initial learning rate set to $3.0e-5$, β_1 and β_2 set to 0.9 and 0.999, respectively, and a mini batch size of 4. The maximum number of iterations was set to 200. Cosine annealing decay is used to adjust the learning rate, the calculation as (23):

$$lr_t = lr_{\min}^i + \frac{1}{2}(lr_{\max}^i - lr_{\min}^i)(1 + \cos(\frac{T_{cur}}{T_i})) \quad (23)$$

where i represents the i times were executed; $lr_{\max}$ and $lr_{\min}$ represent the maximum and minimum values of learning rate, respectively; T_{cur} represents the T_{cur} iteration; T_i represents the number of executions per iteration. The cross-entropy loss function Ross and Dollár (2017) and Lovasz-softmax loss function Berman et al. (2018) are chosen as the

loss functions. The formula of cross-entropy loss function as follows:

$$L_{ce} = - \sum_{i=1}^N y_i \log(p_i) \quad (24)$$

where L_{ce} represents the cross-entropy loss function. The calculation formula of Lovasz-softmax loss as follows.

$$p_i = \frac{e^{y_i}}{\sum_j e^{y_j}} \quad (25)$$

$$m_i = \begin{cases} 1 - p_i, & y_i = 1 \\ p_i, & y_i = 0 \end{cases} \quad (26)$$

$$g_i = \begin{cases} m_i - m_{i-1}, & i > 1 \\ m_i, & i = 1 \end{cases} \quad (27)$$

$$\Delta_i = \begin{cases} \frac{1}{i} \sum_{j=1}^i m_j, & i > 0 \\ 0, & i = 0 \end{cases} \quad (28)$$

$$L_{lov} = \sum_{i=1}^n g_i \cdot \Delta_i \quad (29)$$

where L_{lov} represents Lovasz-softmax loss function. Last, the calculation formula for loss as “(30)”:

$$loss = L_{ce} + L_{lov} \quad (30)$$

The L_{ce} is a commonly used loss function in DL. Essentially, RSCD tasks can be regarded as binary or multi-class classification problems, which involve determining whether each pixel point or region has undergone changes. The cross-entropy loss function can effectively measure the difference between the probability distribution predicted by the model and the actual labels, thereby guiding the model optimization. Compared to L_{ce} , L_{lov} excels in handling class

imbalance issues. It transforms submodular set functions into differentiable convex functions through Lovász extensions, thereby effectively balancing the loss between easily classified samples and difficult ones. This avoids the model's overemphasis on easy examples at the expense of harder ones during training. Combining L_{ce} with L_{lov} can achieve more accurate classification and finer boundary localization in RSCD tasks, especially effective when dealing with datasets that have class imbalance and complex boundary structures. This method helps to improve the overall performance and robustness of the model, particularly in assessing and predicting change areas.

4.3 Evaluation Metrics

For an intuitive comparison of performance, Intersection over Union (IoU), F1 scores (F1), and Overall Accuracy (OA) were selected. The calculation formulas for each metric are as follows.

$$IoU = \frac{TP}{TP + FP + FN} \quad (31)$$

$$F1 = \frac{2 \times TP}{2 \times TP + FP + FN} \quad (32)$$

$$OA = \frac{TP + TN}{TP + TN + FN + FP} \quad (33)$$

where TP represents true positive, TN represents true negative, FN represents false negative, FP represents false positive.

Table 1 Comparisons with SOTA methods on the S2Looking, WHU-CD and GVLM-CD. The best results are indicated in italics.

Methods	S2Looking			WHU-CD			GVLM-CD		
	IoU(%)	F1(%)	OA(%)	IoU(%)	F1(%)	OA(%)	IoU(%)	F1(%)	OA(%)
FC-Siam-diff	39.63	56.76	99.07	68.87	81.57	98.20	72.46	84.03	98.22
BIT	42.36	59.51	99.17	79.68	88.69	98.97	71.82	83.60	98.17
DSAMNet	31.28	47.65	98.49	73.87	84.97	98.60	73.47	84.71	98.28
SNUNet	40.75	57.90	99.04	76.50	86.69	98.80	73.34	83.76	97.66
AERNet	43.33	60.46	99.15	72.94	84.35	98.54	71.87	83.63	98.20
P2V	37.04	54.05	99.06	84.19	91.42	99.25	73.36	84.63	98.36
MSFGNet	38.86	56.13	99.11	82.67	90.24	98.93	73.74	84.89	98.38
SAM-CD	39.60	56.74	99.06	77.83	87.53	98.87	71.49	83.95	97.69
CD-RLKNet	43.84	60.95	99.19	85.21	91.07	99.03	76.21	85.77	98.50
RSMamba	43.72	60.08	99.16	83.46	91.03	99.22	75.81	85.24	98.50
ISDANet	43.17	60.31	99.05	79.72	89.16	98.25	76.35	86.59	98.52
Ours	<i>44.95</i>	<i>62.02</i>	<i>99.23</i>	<i>86.08</i>	<i>92.52</i>	<i>99.35</i>	<i>77.90</i>	<i>87.58</i>	<i>98.98</i>

4.4 Comparison Experiments

To validate the efficacy of LKMamba-CD, this section compares its prediction results with FC-Siam-diff Caye Daudt et al. (2018), BIT Hao Chen and Shi (2021), DSAMNet Shi et al. (2021), SNUNet Fang et al. (2022), AERNet Zhang et al. (2023a), P2V Lin et al. (2023), MSFGNet Huang et al. (2024c), SAM-CD Ding et al. (2024), CD-RLKNet Huang et al. (2024a), RSMamba Zhao et al. (2024a), and ISDANet Ren et al. (2025). Table 1 presents the evaluation metrics for each model.

As shown in Table 1, LKMamba-CD consistently achieves the best results across all three datasets, demonstrating its strong generalization capability and robustness in RSCD tasks. On the S2Looking, LKMamba-CD attains

IoU of 44.95%, F1 of 62.02%, and OA of 99.23%, outperforming CD-RLKNet, second-best, by 1.07% in F1. This improvement highlights the model's enhanced ability to capture subtle land-cover changes in heterogeneous backgrounds. The incorporation of LKS effectively expands the receptive field while maintaining spatial resolution, allowing the model to recognize distributed or small-scale change regions that are often missed by CNN-based methods such as SNUNet and AERNet. On the WHU-CD, LKMamba-CD achieves IoU of 86.08%, an F1 of 92.52%, and OA of 99.35%. Compared with CD-RLKNet and RS-Mamba, the improvements of 1.45% and 1.49% in F1, respectively, demonstrate that the proposed BCIF effectively integrates cross-temporal features, reducing pseudo-change errors. The large receptive field obtained by LKS enables accurate

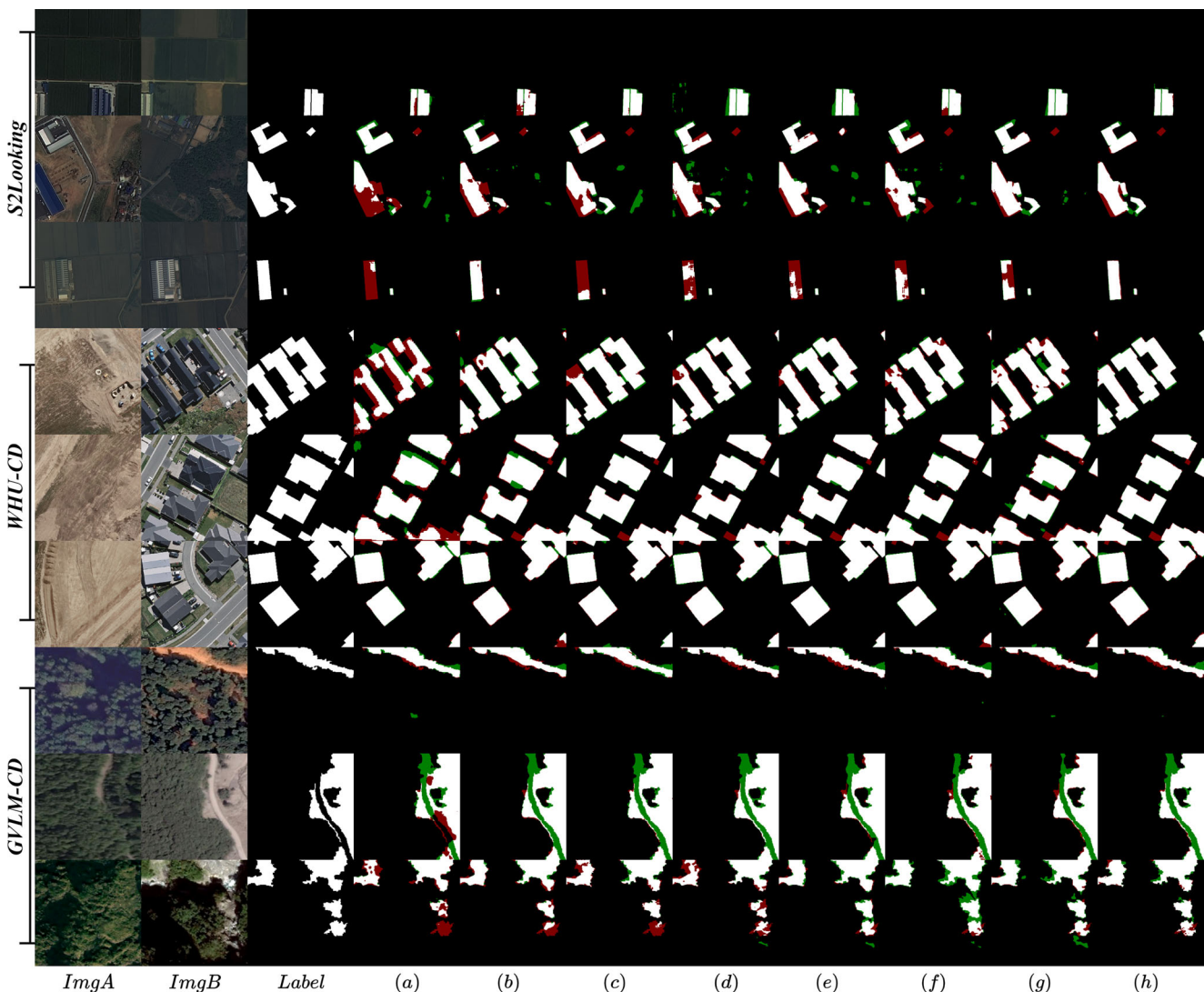


Fig. 4 Visualization results of different change detection methods, where white indicates TP, black indicates TN, green indicates FP and red indicates FN. **a** represents FC-Siam-diff Caye Daudt et al. (2018), **b** represents BIT Hao Chen and Shi (2021), **c** represents SNUNet Fang et al. (2022), **d** represents P2V Lin et al. (2023), **e** represents CD-RLKNet Huang et al. (2024a), **f** represents RSMamba Zhao et al. (2024a), **g** represents ISDANet Ren et al. (2025), and **h** represents LKMamba-CD (Ours)

Table 2 Comparison results of computational cost.

Methods	Params.(M)	Flops(G)	VRAM(G)	Infer.(ms)
FC-Siam-diff	1.35	5.31	0.18	1.93
BIT	3.50	10.63	0.49	2.81
DSAMNet	16.95	47.24	0.67	4.15
SNUNet	10.21	46.36	0.20	4.79
AERNet	25.36	12.82	0.43	2.89
P2V	45.43	32.75	0.65	6.93
MSFGNet	12.89	17.24	0.31	3.16
SAM-CD	70.59	8.60	0.48	2.39
CD-RLKNet	31.24	47.22	0.20	4.57
RSMamba	42.32	18.36	0.38	2.04
ISDANet	6.40	3.80	0.45	1.75
Ours	8.20	3.38	0.27	1.52

extraction of global structural context, while the LRFA module further enhances spatial continuity in reconstructed change maps. For the GVLM-CD LKMamba-CD achieves IoU of 77.90%, an F1 of 87.58%, and an OA of 98.98%, surpassing ISDANet, the second best, by 0.99% in F1. This dataset is particularly challenging due to large-scale environmental differences and low spectral consistency. The superior results of LKMamba-CD confirm that the joint use of large-kernel convolutions and state-space modeling enables the network to balance local texture preservation and global semantic coherence, achieving both boundary precision and contextual awareness.

Overall, these results demonstrate that LKMamba-CD successfully combines the advantages of convolutional inductive bias and global dependency modeling, offering a scalable and accurate solution for large-scale RSCD applications. The significant gains across all datasets underline the effectiveness of its hybrid architecture, which achieves a favorable balance between spatial detail preservation, global context understanding, and computational efficiency. To visually compare the performance of different models, the prediction results of some models are illustrated in Fig. 4.

Figure 4 shows that the LKMamba-CD's predictions are closest to the ground truth for change detection across all datasets. LKMamba-CD not only maintains a leading position across all datasets but also exhibits a notable margin of superiority, showcasing its robust generalization capability and exceptional performance RSCD tasks. It can more accurately identify areas changed, improve the overall classification accuracy, and provide more reliable technical support for practical applications. Specifically, the prediction results on S2Looking highlight a significant misclassification by other models, as evidenced by the numerous red and green regions. However, while LKMamba-CD does not achieve perfect accuracy, its performance is notably superior. Although some models provide satisfactory results in

certain scenarios, LKMamba-CD consistently outperforms them in accurately identifying the changed areas. The visualization results of WHU-CD indicate that the performance of each model aligns with the trends observed in the metrics presented in Table 1, showcasing distinct features in detecting the location and category of change objects in the images. For GVLM-CD, most models tend to overlook differences in land cover and focus more on shallow information, a trend particularly evident in the sixth row of Fig. 4. However, LKMamba-CD accurately identifies the unchanged portion in the upper right corner, demonstrating its superiority over other models.

4.5 Efficient Comparison

To compare the computational efficiency of different models, we evaluate their computational costs on the GVLM-CD dataset, including number of parameters (Params.), floating-point operations (Flops), peak video RAM (VRAM), and per-image inference time (Infer.) Table 2 present the efficiency metric.

As demonstrated in Table 2, LKMamba-CD exhibits exceptional computational efficiency, achieving the lowest Flops of 3.38G and the fastest Infer. of 1.52ms among all evaluated methods. Comparison with CNNs-based methods, CD-RLKNet effectively expands the receptive field, it incurs a massive computational burden with Flops of 47.22G. In contrast, LKMamba-CD leverages structural re-parameterization within the LKS module, significantly expanding the receptive field during training while absorbing the multi-branch depthwise convolutions into a single kernel during inference, thereby accelerating computation without bloating Flops. Comparison with Transformers-based methods, BIT and SAM-CD require considerable computational resources. Moreover, the recent SSM-based RSMamba requires 42.32M parameters and 18.36G Flops. LKMamba-CD, conversely, only requires 8.20M parame-

rately detect changed objects. Meanwhile, it is also noted that as the kernel size gradually increases, the performance improvement gradually diminishes. Therefore, the size of *RepLK* in LKS is set to 13.

The removal of BCIF results in a noticeable drop in F1 by 4.84%, 6.56%, and 4.40% on the S2Looking, WHU-CD, and GVLM-CD datasets, respectively. This decline underscores BCIF's crucial role in capturing and aligning cross-temporal semantics. By leveraging the LKS to model long-range dependencies between bi-temporal features, BCIF effectively distinguishes true changes from illumination or seasonal variations. It enhances the model's robustness against pseudo-changes, particularly in datasets with strong environmental variability such as S2Looking. Similarly, omitting LRFA leads to a marked reduction in detection accuracy, as F1 decreases by 3.26%, 6.66%, and 3.80% on the three datasets. This result demonstrates that LRFA contributes significantly to refining high-level semantic representations and preserving structural consistency in change regions. By incorporating both efficient attention and LKS, LRFA enhances the aggregation of spatially distributed change features across multiple scales, thereby improving boundary precision and mitigating fragmented predictions.

In addition, it can be observed that in the RSCD task, the performance of using LKS is superior to that of using VSS-Block. Similarly, within LKS, the use of *RepLK* demonstrates better performance compared to regular convolution. As the kernel size of the *RepLK* increases, the model's performance also improves, but the rate of improvement becomes increasingly slower. To intuitively compare the impact of different modules on model performance, Fig. 5 presents the visualization results of the ablation study.

Figure 5 shows that the LKMamba-CD in (f) exhibits the best performance. When comparing (a), (b) and (f), it is evident that the model performance deteriorates significantly in the absence of BCIF and LRFA, resulting in severe FN. This indicates that BCIF and LRFA are capable of extracting and fusing remote sensing difference information, thereby effectively enhancing the model's performance. Column (c) utilizes VSSBlock replace LKS, the visualization results suggest that LKS outperforms VSS-Block in terms of performance. In summary, the ablation experiments confirm that the choice of kernel size and the inclusion of BCIF and LRFA significantly affect the performance of LKMamba-CD. The optimal configuration, as demonstrated by the results, combines a larger kernel size with both modules to achieve best performance, showcasing the model's robustness and effectiveness in change detection tasks.

To explicitly demonstrate the trade-off between performance and computational cost in the selective scanning mechanism, we evaluated the impact of different scanning strategies within the LKS. As shown in Table 4, we compared 1D horizontal scan mechanism (1D-H), 1D vertical scan mechanism (1D-V), 2D scan mechanism (2D-HV), and SS2D (Ours). In addition, In Table 4, we compare different scanning mechanisms in terms of Params.(M), Flops(G), VRAM(M), and Infer.(ms) on the GVLM-CD dataset. The results highlight a highly favorable trade-off for the four-scan strategy. While the 1D scan strategies (1D-H/V) require the lowest memory consumption of 227.61 M and computational time of 1.47 ms, they severely limit the SSM's ability to model non-causal spatial dependencies, leading to a noticeable degradation in detection accuracy. Conversely, Ours achieves the highest F1 of 62.02%, 92.52%, and 87.58% on S2Looking, WHU-CD,

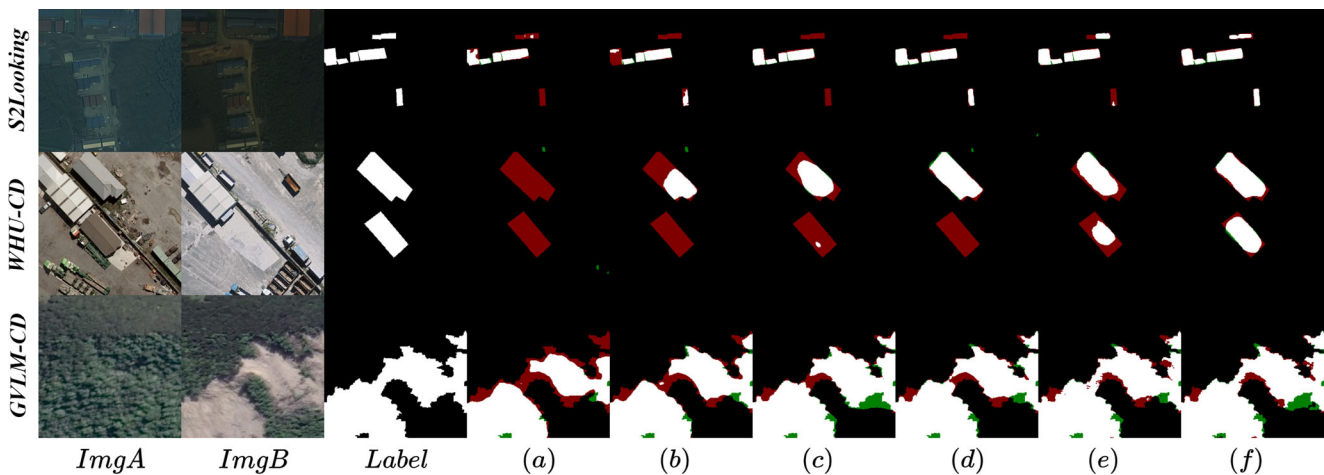


Fig. 5 Visualization results of ablation experiments, where white indicates TP, black indicates TN, green indicates FP and red indicates FN. **a** represents BCIF is absent, **b** represents LRFA is absent, **c** represents VSS, **d** represents kernel size of 3, **e** represents kernel size of 9, and **f** represents kernel size of 13 (LKMamba-CD)

Table 4 Ablation on scanning strategies on various datasets.

	Params.(M)	Flops(G)	VRAM(M)	Infer.(ms)	S2Looking		WHU-CD		GVLM-CD	
					IoU(%)	F1(%)	IoU(%)	F1(%)	IoU(%)	F1(%)
1D-H	8.12	3.14	227.61	1.47	40.07	57.22	81.60	89.87	73.40	84.66
1D-V	8.12	3.14	227.61	1.47	40.10	57.24	81.54	89.93	73.61	84.80
2D-HV	8.15	3.32	262.29	1.50	41.51	58.67	84.74	91.74	76.28	86.54
Ours	8.20	3.38	268.95	1.52	44.95	62.02	86.08	92.52	77.90	87.58

and GVLM-CD, respectively. More importantly, this significant performance leap comes at a negligible computational cost, compared to the 1D scan mechanism, ours methods merely adds 0.24G Flops, 41.34M in peak VRAM, and 0.05ms in inference time. This confirms that the proposed architecture successfully achieves an optimal balance between high-precision change detection and computational efficiency.

5 Discussion

The experimental results comprehensively demonstrate that the LKMamba-CD achieves superior performance compared with existing RSCD methods. The effectiveness of LKMamba-CD can be attributed to several key innovations. LKS couples RepLK with SS2D, allows for expansive receptive fields that capture subtle, large-scale variations in land cover, such as urban expansion or disaster-induced alterations, without introducing excessive noise or computational overhead. While LKMamba-CD demonstrates superior overall performance, a balanced evaluation of its operational boundaries is crucial for identifying ongoing challenges in RSCD. Figure 6 visualizes representative failure cases, illustrating extreme conditions where the model fails to segment change regions correctly.

A fundamental operational boundary of LKMamba-CD lies in the inherent trade-off between the expansive receptive field and the preservation of fine-grained spatial details. While our architecture mitigates excessive smoothing by integrating varying dilation rates within the RepLK and preserving high-resolution (1/8 scale) shallow features through the BCIF, extreme micro-changes remain highly challenging. As explicitly visualized in Fig. 6 (S2Looking, Pair I), LKMamba-CD occasionally produces significant FN when encountering extremely small, sparse, and scattered building changes. This visually confirms that the large RF, while excellent for extracting macroscopic context, inherently acts as a spatial low-pass filter. When target changes are disproportionately small relative to the global context, the dominant large-scale feature representations may suppress these subtle variations, “smoothing out” the fine-grained, pixel-level micro-changes. This indicates that applications requiring microscopic precision may require specialized high-frequency feature enhancement techniques in future research.

In very high-resolution urban scenarios, LKMamba-CD sometimes fails to detect the complete spatial extent of massive newly constructed facilities. As seen in Fig. 6 (WHU-CD, Pair I), LKMamba-CD successfully delineates the boundaries of the new massive factory but yields large FN in the central roof area due to extreme textural homogeneity. Additionally, highly chaotic semantic environments, such as active construction sites with temporary

materials and cranes, such as Pair III, introduce complex, high-frequency visual noise that confuses the feature representations, leading to intertwined False Positives and False Negatives. Detecting complex geological events like landslides presents a unique challenge. In Fig. 6 (GVLM-CD, Pairs I and II), LKMamba-CD correctly identifies parts of the landslide but misses significant portions of the topological change. This failure occurs because the spectral signature of the newly exposed soil blends almost perfectly with the surrounding pre-existing bare land or sparse vegetation. These failure cases underscore that relying solely on bi-temporal RSIs possesses inherent limitations. While LKMamba-CD pushes the boundaries of 2D spatio-temporal modeling, fully resolving these edge cases—particularly spectral blending and severe geometric distortions motivates the future integration of multi-modal data to achieve more robust, physics-aware environmental monitoring.

To address these operational boundaries, our future work will focus on three key directions. First, to overcome the visual limitations of bi-temporal optical imagery, we aim to extend the architecture for multi-temporal and multi-source data integration. Second, we will explore self-supervised and few-shot learning strategies to alleviate the reliance on large-scale annotated datasets. Finally, to further enhance scalability for global, near-real-time monitoring, we plan to incorporate model compression and knowledge distillation techniques to strictly minimize computational overhead.

6 Conclusion

DL-based land cover change detection is one of the key research directions in the field of RSCD. To obtain the correlation of large-scale and long-range land cover features, this work introduces LKMamba-CD that combines large-kernel convolution and Mamba to achieve efficient

and accurate RSCD. LKMamba-CD utilizes UniRepLKNet to extract multi-dimensional land cover features. Within LKMamba-CD, LKS combines re-parameterized large kernel convolution with Mamba to capture larger receptive fields and long-range feature correlations. In addition, BCIF and LRFA are designed using LKS as the core component. BCIF is designed to extract the difference information of land cover features at different temporal, thereby integrating the change information of multidimensional spatio-temporal features. LRFA uses efficient attention and LKS to strengthen the correlation of change features in different dimensions through large-scale feature extraction, allowing for the preservation of high-dimensional semantic features without losing large-scale change information across different dimensions. Through experimental validation, LKMamba-CD has shown strong performance in the S2Looking, WHU-CD, and GVLM-CD datasets, achieving F1 scores of 62.02%, 92.52% and 87.58%, respectively. LKMamba-CD achieves an optimal balance between spatial precision and global dependency modeling, which is critical for large-scale and complex land-cover change analysis.

These results indicate the effectiveness of LKMamba-CD in accurately detecting land cover changes across different datasets, highlighting its potential for use in remote sensing applications. The combination of re-parameterized large kernel convolution with Mamba in LKMamba-CD enables the extraction of larger receptive fields and long-sequence feature correlations, which contributes to the improved performance observed in these experiments. Future research will focus on extending the architecture for multi-temporal or multi-source remote sensing data integration, addressing real-world demands beyond bi-temporal analysis and incorporating self-supervised or few-shot learning to minimize reliance on large labeled datasets, enhancing scalability.

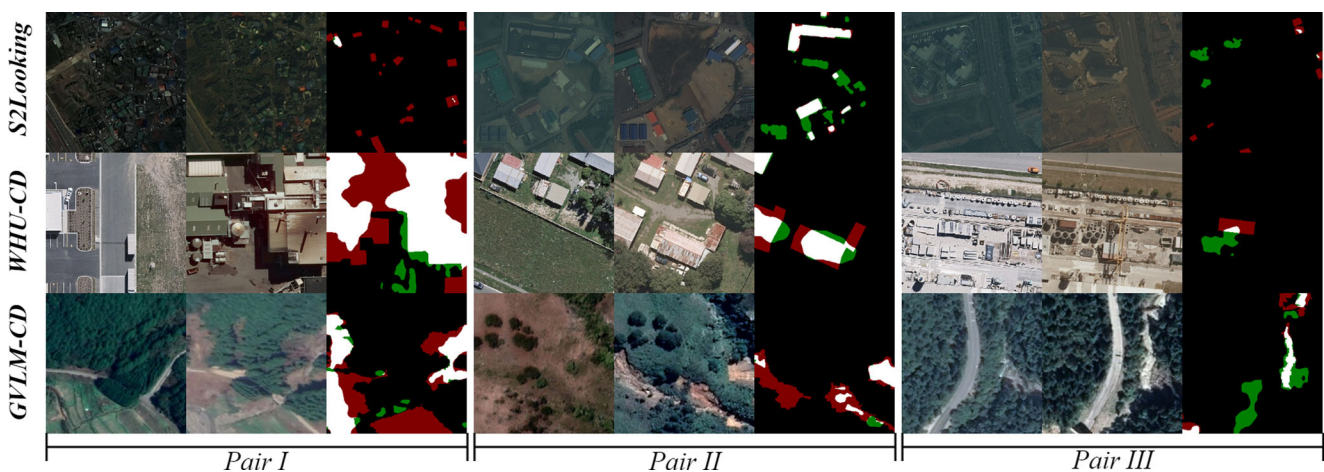


Fig. 6 Samples of limitation and failure case, where white indicates TP, black indicates TN, green indicates FP and red indicates FN

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Data Availability No datasets were generated or analysed during the current study.

Declarations

Competing interests The authors declare no competing interests.

Consent for publication Not applicable.

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